# West University of Timișoara: Machine Learning Systems for Sleep Quality Assessment Based on EEG Signals


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## Introduction

**Context**: Sleep disorders affect a significant percentage of the global population, severely impacting quality of life and long-term health.
**Limitations**: Standard polysomnography (PSG) is invasive, expensive, and manual scoring of sleep stages is time-consuming and prone to human subjectivity [1].
**Proposed solution**: Automate sleep stage analysis (Wake, REM, N1, N2, N3) using a reduced number of sensors (single EEG channel) and artificial intelligence algorithms.

## Objectives

1. Development of a complete pipeline for preprocessing and automatic analysis of raw EEG signals.
2. Identification and extraction of the most relevant time and frequency domain features for brain
3. Training and comparing the performance of multiple Machine Learning models for accurate hypnogram generation.

waves.

## Proposed Methodology

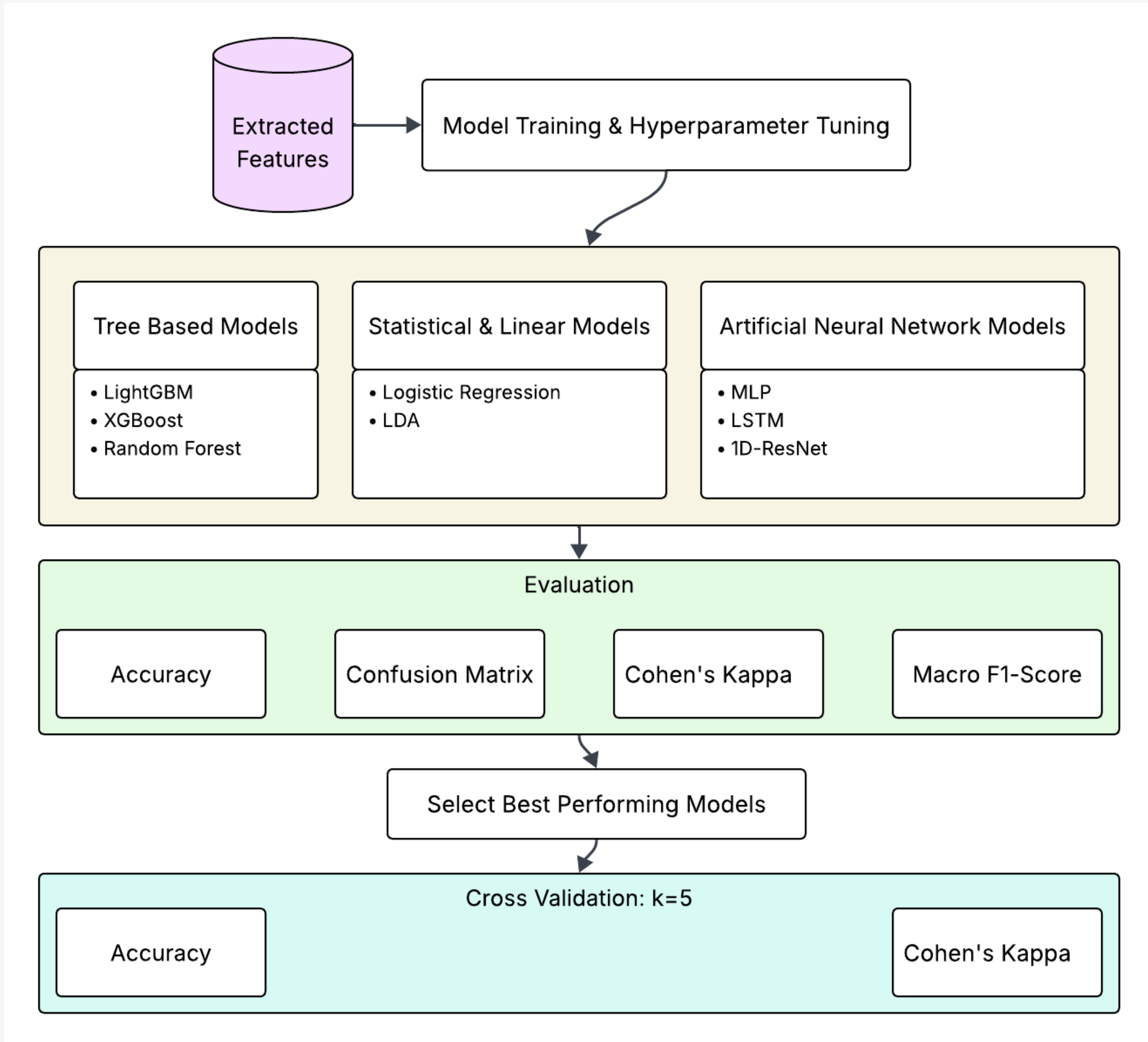
**Dataset**: Three of the most widely used open-source datasets in this field were used: Sleep-EDF Expanded, UCD Apnea and HMC Sleep Stage [2, 3, 4].

**Data Engineering**: Due to the heterogeneity of the selected studies, it was necessary to implement a rigorous data transformation step. The workflow included quality validation, epoch segmentation, elimination of non-sleep epochs, and standardization of signals and annotations in a common format, a mandatory condition for training the models.

**Preprocessing and Feature Extraction**: EEG signals were filtered and standardized (Z-Score per subject) to reduce individual differences in cranial architecture. The extraction involved the transfer of the EEG signal into numerical variables, integrating time-domain and frequency-domain methods:

- Physiological Event Detection - specific isolation of sleep spindles (11-16 Hz) and slow waves (0.5-2 Hz).
- Hjorth Parameters - Mobility and Complexity were extracted because they serve as robust markers for rapid identification of transitions between sleep stages [1].
- Fine Spectral Structure - to measure the similarity of the spectrum to noise and to describe the general shape and energy distribution of the waves.
- Contextual Engineering - To capture the cyclical nature of sleep, clinical reports were generated, and rolling windows (2 epochs forward, 2 epochs backward) were applied.

**Models used**: The classification models, divided into tree-based, probability and statistics, and neural networks, were fully tested on the extracted features.




Models and Evalutation

## Results

| Model | Mean Acc (%) | Mean Kappa | Best Acc (%) | Best Kappa |
|---|---|---|---|---|
| LightGBM | 77.11 | 0.6888 | 78.58 | 0.7113 |
| XGBoost | 77.04 | 0.6880 | 78.36 | 0.7086 |
| Random Forest | 74.33 | 0.6507 | 76.02 | 0.6763 |
| LDA | 72.51 | 0.6276 | 73.91 | 0.6459 |
| Logistic Regression | 71.21 | 0.6258 | 72.99 | 0.6477 |
| MLP | 77.61 | 0.6988 | 78.95 | 0.7182 |
| LSTM | 77.4 | 0.6943 | 77.7 | 0.6995 |
| ResNet | 77.47 | 0.6965 | 77.89 | 0.7008 |

Model performance

## Propose Architecture

As can be seen in Table 1, the best results were given by the following two architectures:

- Multilayer Perceptron (MLP)
  - A *feedforward* neural network structured on three dense layers (256, 128 and 64 neurons), designed to gradually condense the extracted features into high-precision nonlinear representations.
  - Integrates *Batch Normalization* layers to accelerate convergence and stabilize training, along with *Dropout (30%)* mechanisms strategically applied to prevent overfitting.
- Residual Network (ResNet)
  - Uses convolutional blocks (Conv1D) with 32, 64 and 128 filters to identify specific micro-events in the EEG signal.
  - Implements *shortcut/skip-connection* mechanisms to facilitate gradient propagation and allow efficient training of a deeper network.
  - Uses *Global Average Pooling (GAP)* before the classification layer to reduce spatial dimensionality and increase the robustness of the model to noise variations.

## Conclusions

The use of a single EEG channel combined with advanced feature extraction represents a viable and accurate alternative to traditional multi-channel analysis.
Deep learning models are able to efficiently capture the complex and nonlinear relationships specific to brain activity during sleep.

## Future Directions

- Creating models trained on raw data and testing if the variant is less expensive in terms of execution time and generating histograms.
- Optimization of the code for real-time analysis and integration of the algorithm into wearable devices.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[4] D. Alvarez-Estevez and R. Rijsman, "Haaglanden Medisch Centrum Sleep Staging Database v1.1," 2022. Accessed: 2026-03-08.